

Introduction to Operating System Administration

Tasks of the Sysadmin

René Serral-Gracià¹

¹Universitat Politècnica de Catalunya (UPC)

February 16, 2026

Subject Topics

- ❶ **Operating System Administration Introduction**
- ❷ Operating System installation
- ❸ User management
- ❹ Application management
- ❺ System monitoring
- ❻ Filesystem Maintenance
- ❼ Network services
- ❽ Security and Protection
- ❾ Introduction to Public Cloud

Outline

- 1 Introduction
- 2 Sysadmin tasks
- 3 Administrator's ethical code
- 4 Base concepts

Outline

- 1 Introduction
 - Goals
- 2 Sysadmin tasks
- 3 Administrator's ethical code
- 4 Base concepts

Goals

- Understand what a Sysadmin is
 - Tasks
 - Responsibilities
 - Necessary knowledge
- Understand and honor the Sysadmin ethical code

Outline

- 1 Introduction
- 2 Sysadmin tasks**
- 3 Administrator's ethical code
- 4 Base concepts

Exercise

Sysadmin tasks

- List the sysadmin tasks
 - Which of those activities are you able to perform?
- Gather the task list
 - Sort them by importance

Solution

- Check system's security
- Perform regular backups
- Solve incoming issues
- Assist end-users in their daily work
 - Related to their systems
- Keep local documentation
- Install and manage software
- Manage users
- Add/update hardware
- Monitor the system and ensure it works properly

Outline

- 1 Introduction
- 2 Sysadmin tasks
- 3 Administrator's ethical code**
- 4 Base concepts

Exercise

In group

- Has the Sysadmin the right to read e-mails from other users?
- And to monitor which web pages they visit?

Exercise

In group

- Has the Sysadmin the right to read e-mails from other users?
- And to monitor which web pages they visit?

Example

An employee is fired from a company because he/she spent too much time sending personal e-mails to his/her friends irrelevant to the company.

- Which is the role of the Sysadmin in this situation?

Administrator ethical code

Solution

- The Administrator ethical code forbids to access the information contained in the users e-mails
He/she is not allowed to monitor by his/her own, it must be “company policy” and the users must be informed that their data might be monitored
- Has provided the information as requested by any of his employers
- If employee contracts say that the Intellectual Property generated in the company belongs to them. Then the company can legally act against an employee using company's resources for personal benefit

Administrator ethical code

Definition (by Gemini)

Being a sysadmin is a bit like being a digital locksmith with a master key; it requires a high level of integrity because you have the power to see almost everything

Administrator ethical code

- Professionalism
 - **Maintain Integrity:** Do not use your access for personal gain or to settle scores
 - **Take Responsibility:** Own your mistakes



If a script wipes a partition, be the first to report it and the first to start the restore

- **Keep Learning:** Technology moves fast



Staying current is considered an ethical obligation to ensure the security and efficiency of the systems you manage

Administrator ethical code

- Personal Integrity and Privacy
 - **Respect Confidentiality:** This is the big one

 You will see private emails, sensitive files, and HR data. You must treat this information as "privileged" and never access it without a specific, documented business need
 - **Avoid Conflicts of Interest:** Don't accept kickbacks from vendors or engage in activities that compromise your duty to your employer
- Laws and Policies
 - **Educate Yourself:** Understand the local laws (like GDPR or NIS2) and your company's internal policies
 - **Adherence:** Follow the rules, even when they make your job slightly more difficult

Administrator ethical code

- Communication and Cooperation
 - **Share Knowledge:** Don't build a "kingdom of one" by hoarding information or refusing to document your work
 - **Professional Conduct:** Treat users, colleagues, and management with respect, even when they ask for the tenth time how to reset their password
- System Integrity
 - **Security First:** You are the gatekeeper

 It is your duty to ensure the system is protected against unauthorized access and data loss
 - **Reliability:** Design systems for uptime and recoverability

Administrator ethical code

- Education, laws and usage policies
 - Be up-to-date in terms concerning ethical norms used in this/her work, share the knowledge with others
 - Criminal code, LOPD, LSSI
- Communication
 - Inform the users about important data
- Ethical responsibility
 - Proficiency, coherency, integrity and honesty

LOPD (Data protection and Organic law)

- Determines the treatment of data files (databases)
 - Name/surname, ID, direction, bank accounts, credit card numbers, . . .
- The company must inform about the presence of files to the Data Protection Agency
- The company must adopt determined security measures depending on the privacy of the data file
- The company must write a document clearly stating the taken security measures
- The employees' contracts must be coherent with the law in terms of the stored private information

You must be up-to-date!!

Outline

- 1 Introduction
- 2 Sysadmin tasks
- 3 Administrator's ethical code
- 4 Base concepts**

Base Concepts for next class

- UNIX knowledge at user level
 - `ls`, `cd`, `mkdir`, `rmdir`, `cp`, `mv`, `rm`, `cat`, `more`
 - **Shell**, redirection operators (`<`, `>`, `&`), and pipes (`|`)
 - `apropos`, `man`
 - `grep`, `find`
 - `vi`
- Hard disk preparation and partitioning
 - `fdisk`, `parted`, `mkfs`, `mkswap`, `fsck`, `mount`